PROJECT 1

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1. Network Type Assessment: Discuss the benefits and constraints of different network types and standards.

1.Local Area Network (LAN)

LANs work well for linking devices in a small area, like an office or a building. They use Ethernet for wired connections and Wi-Fi for wireless ones.

Benefits

- Fast data transfer (up to 10 Gbps).
- Affordable setup for local use.
- Easy management and problem-solving with centralized control.
- Can easily add more devices or expand the network.

Constraints

- Limited to one building or location.
- Centralized systems can have weak spots.
- Wireless LANs can be accessed by unauthorized users.
- Wired LANs can be costly and complicated to set up in big spaces.

2. Wide Area Network (WAN)

Wide Area Networks (WANs) link several Local Area Networks (LANs) across big areas using methods like MPLS, SD-WAN, or rented lines.

Benefits

- Allows offices or remote sites to connect worldwide.
- Works well with cloud services.
- Can grow with the company.
- Gives easy access to resources spread out over different locations.

Constraints

- Expensive to set up and maintain because of rented lines and telecom fees.
- Can have slow speeds over long distances.
- Needs skilled people to set up and manage.
- Vulnerable to security risks unless protected by strong measures like VPNs.

3.Metropolitan Area Network (MAN)

MANs link several LANs in a city or campus, usually using fiber-optic cables for fast connections.

Benefits

- Covers a large area, good for big campuses or cities.
- Fast data transfer with modern fiber-optic technology.
- Affordable for connecting many places in a city.
- Reliable for big companies with multiple offices nearby.

Constraints

- Limited to medium-sized areas (like cities).
- Depends on service providers for setup and help.
- Not ideal for worldwide connections.
- Problems in the main cable can affect many connected places.

4. Wireless Networks (WLAN/WPAN)

Wireless networks don't need cables, which makes it easier to move devices around. WLANs work over large areas, while WPANs, like Bluetooth, are for short distances.

Benefits

- Setting up is faster and simpler than wired networks.
- Users can move around while staying connected.
- Adding more devices is easy without needing extra cables.
- It's cheaper for offices that change often.

Constraints

- Other devices or objects can cause problems with the connection.
- It's slower than wired connections.
- There are risks like people listening in or getting access without permission.
- Regular updates are needed to keep it working well and up-to-date.

5. Virtual Private Network (VPN)

VPNs provide secure remote access to private networks using encrypted connections over the internet.

Benefits

- Ensures secure and private communication, protecting sensitive data.
- Cost-effective alternative to leased lines for remote connectivity.
- Enables employees to work remotely without compromising security.
- Flexibility to connect multiple locations or users.

Constraints

- Performance can be affected by encryption overhead and internet quality.
- Misconfiguration can lead to security vulnerabilities.
- Latency may impact real-time applications like video conferencing.
- Requires robust infrastructure and ongoing monitoring to prevent breaches.

Physical Layer Standards

Benefits

- Standardized hardware (e.g., Ethernet cables, USB) ensures compatibility between devices from different manufacturers.
- Widespread adoption of standards like IEEE 802.3 (Ethernet) lowers production costs through economies of scale.
- Standards ensure consistent performance in physical connections and data transfer rates.

Constraints

- Older standards like Cat5 Ethernet cables may not support high-speed data transfer required by modern applications.
- Upgrading to newer standards (e.g. Cat6a cables or fiber optics) can be expensive for organizations.
- Distance and interference limits (e.g. signal degradation in wired and wireless connections) are dictated by the standards.

Transport Layer Standards

Benefits

- TCP ensures error-free, ordered delivery of data.
- Protocols like UDP allow faster communication for real-time applications (e.g.gaming, VoIP).
- Standards support various use cases, from streaming to secure communications.

Constraints

- TCP's reliability mechanisms increase overhead, reducing efficiency for small or time-sensitive data.
- Transport layer standards (e.g. UDP) lack built-in encryption or authentication, requiring additional layers for security.
- Choosing between TCP and UDP depends on application requirements, limiting flexibility in certain scenarios.

2.Network Topology: Explain the impact of network topology, communication and bandwidth requirements.

How does the choice of network topology impact the efficiency of communication between devices and the overall bandwidth requirements of the network?

The choice of network topology directly impacts the efficiency of communication between devices by determining how data travels through the network, how devices are connected, and how the network handles traffic.

1. What Requires to Enhance Bandwidth?

Enhancing bandwidth involves improving the capacity to transmit data across a network. Several factors contribute to enhancing bandwidth

Upgrading Infrastructure

- Replacing copper cables with fiber optic cables can increase the bandwidth since fiber offers much higher data transmission rates and lower latency.

- Modern network equipment, such as high-performance routers, switches, and network interface cards (NICs), can handle more data and allow for faster communication between devices.

Improving Network Configuration

- Combining multiple network links (e.g., using LACP - Link Aggregation Control Protocol) can effectively increase available bandwidth between devices by utilizing several links simultaneously.

- Controlling the flow of data in the network to prioritize important traffic and manage less critical traffic can optimize bandwidth usage.

Wireless Optimization

- Wi-Fi 6 (802.11ax) The latest wireless technology supports higher speeds and better performance in environments with multiple devices.

- Reducing interference in wireless networks by choosing appropriate channels and ensuring minimal overlap can enhance bandwidth utilization.

Reducing Network Congestion

- Distributing network traffic evenly across multiple servers or pathways helps prevent bottlenecks and ensures a smoother flow of data.

- Prioritizing certain types of traffic (e.g., VoIP, video conferencing) ensures that critical services receive adequate bandwidth even during high traffic periods.

2. Bandwidth Requirements for Different Topologies

Star Topology

Bandwidth Requirements

- In a star topology, each device connects directly to a central hub or switch. The total bandwidth available to the network depends on the capability of the central hub and the individual connections between the devices and the hub.

- Higher Bandwidth is needed for the central hub as it becomes the point of communication for all devices. If the hub is overloaded, it could become a bottleneck, limiting bandwidth availability.

Advantages

- Simple to expand devices can be added without affecting the rest of the network.

- Fault tolerance is high if one device fails, others remain unaffected.

Disadvantages

- The central hub can become a bottleneck if it lacks sufficient bandwidth to handle multiple devices.
- The cost of maintaining the hub increases with the number of devices.

Bus Topology

Bandwidth Requirements

- Bus topology relies on a single central communication line (the bus) shared by all devices. All devices use the same channel for communication.
- Lower Bandwidth is available in bus topology because all devices are competing for the same communication path, which can cause congestion.
- Bandwidth is shared among devices, and performance degrades as more devices are added to the bus.

Advantages

- Easy to install and extend, with minimal cable requirements.
- Low-cost setup for small networks.

Disadvantages

- Difficult to scale; as more devices are added, the network becomes slower due to bandwidth sharing.
- If the bus fails, the entire network goes down.

Ring Topology

Bandwidth Requirements

- In ring topology, each device is connected to two other devices, forming a ring. Data travels in one direction (or two, in a dual-ring setup).
- Bandwidth usage depends on the speed of the network and the number of devices. Since data circulates in the ring, all devices share the bandwidth, leading to potential delays if the network is overloaded.

Advantages

- Simple and efficient for smaller networks.
- Easier to troubleshoot because of its predictable data path.

Disadvantage

- A failure in any single device or connection can bring down the entire network.
- Performance can suffer as more devices are added, as each device has to wait its turn to transmit data.

Mesh Topology

- **Bandwidth Requirements**
- Mesh topology connects each device directly to every other device. This results in multiple communication paths.
- High Bandwidth Mesh networks can handle large amounts of data and prevent bottlenecks since data can travel through various paths, balancing the load.
- This topology is particularly suitable for high-demand environments like data centers, where large bandwidth is required for high availability and reliability.

Advantages:

- High fault tolerance even if one link fails, data can take alternative routes.
- Very scalable and efficient for large-scale networks.

Disadvantages

- Expensive to set up and maintain due to the large number of connections.
- More complex to manage and troubleshoot, especially as the network grows.

Impact on Communication Efficiency and Bandwidth

Communication Efficiency

- Star topology offers efficient communication, as data only needs to pass through a central hub, but the hub must have enough capacity.
- Bus and Ring topologies can result in delays and inefficiencies as more devices are added and the bandwidth becomes more saturated.
- Mesh topology offers the best efficiency, as multiple paths ensure uninterrupted and fast communication, even if one path fails.

Bandwidth Allocation

- In star and mesh topologies, bandwidth is more easily allocated because each device typically has a dedicated connection to the central hub or directly to other devices.
- Bus and ring topologies require careful management of shared bandwidth to avoid congestion and ensure adequate performance.

Bottlenecks and Traffic Management

Bottlenecks

- Star topology can experience bottlenecks at the central hub, especially if many devices are communicating at once.
- Bus topology is more prone to bottlenecks as every device shares the same communication channel, leading to slowdowns as more devices are added.
- Ring topology can suffer from bottlenecks if there is a lot of data circulating in the network, especially in larger networks.
- Mesh topology has fewer bottlenecks due to multiple paths for data transmission, though it requires more sophisticated traffic management to avoid excessive use of certain paths.

3.Compare common networking principles and essential protocols and analyze how these will ensure efficient system operation.

What are the Common Networking Principles

Common networking principles are foundational concepts that guide the design and operation of networks, ensuring they are efficient, secure, and scalable. Data transmission involves sending data across networks, typically using methods like packet switching or circuit switching. Redundancy ensures reliability by providing backup paths or systems in case of failures. Scalability allows networks to grow and adapt to increasing traffic or devices without significant performance loss. Security protects data and systems from unauthorized access through encryption, firewalls, and access controls. Interoperability ensures that devices from different vendors can work together by adhering to standard protocols. Performance optimization focuses on minimizing latency and maximizing throughput, often using techniques like QoS and bandwidth management. Fault tolerance ensures the network remains functional during failures, often through redundant hardware and resilient design. Other principles include addressing and routing, resource sharing, modularity, and usability, all of which contribute to network efficiency and effectiveness.

what are the Networking Protocols

Networking protocols are standardized rules that define how devices communicate and exchange data over a network. TCP/IP ensures reliable data transmission by establishing connections and managing addressing and routing. HTTP/HTTPS is used for web content transfer, with HTTPS adding encryption for secure communication. DNS translates domain names into IP addresses, allowing devices to locate each other. SMTP handles email sending, while IMAP/POP3 manage email retrieval, with IMAP allowing synchronization across devices and POP3 downloading emails for offline use. FTP/SFTP facilitates file transfers, with SFTP adding encryption for security. ARP maps IP addresses to MAC addresses within local networks, and DHCP automates IP address assignment. ICMP provides error messaging and operational information, commonly used in tools like ping. SNMP monitors and manages network devices, while BGP is crucial for routing data between different networks on the internet. These protocols ensure reliable, secure, and efficient network communication.

How These Principles and Protocols Work Together

1. Data Transmission with TCP/IP. Data is transmitted using TCP/IP, ensuring it reaches its destination reliably and in the correct order. If there are issues (like packet loss), TCP handles retransmissions to maintain the integrity of the transmission. This principle of data transmission is fundamental to ensuring that any type of data whether emails, websites, or videos can be sent and received without errors.
2. Redundancy with TCP/IP and DNS. Redundancy is achieved in networking through multiple routing paths (IP) and backup servers (DNS). This ensures that if one part of the network fails (like a server or path), the data can still reach its destination through an alternative route or server. For example, DNS often has multiple server replicas worldwide, so if one server is down, another can take its place.
3. Scalability with TCP/IP and DNS. As networks grow, both TCP/IP and DNS can handle the increased load. TCP/IP's routing system allows the internet to grow by adding more routers and servers without disrupting traffic. Similarly, DNS servers are distributed globally, making it easy to scale the internet and handle millions of requests from users worldwide.
4. Security with Protocols. Security protocols like HTTPS (which is HTTP over SSL/TLS) ensure that the data transmitted across the network is encrypted and secure. Redundancy also plays a role in security by providing failover paths for data in case of cyberattacks or network failures.

The comparison between common networking principles and essential protocols highlights how each contributes to the efficient operation of networked systems in different ways.

Data Transmission vs. TCP/IP

Data transmission is a fundamental principle that ensures data moves efficiently across a network. This principle involves methods like packet switching or circuit switching, where information is broken down into packets and routed through networks. Efficient data transmission ensures that information reaches its destination without unnecessary delays. TCP/IP protocols provide the framework for reliable data transmission. TCP ensures that data packets are delivered in the correct order, retransmitting any lost packets, while IP manages the routing of data packets between devices using unique IP addresses. Together, they enable the reliable and efficient flow of data over the internet.

Redundancy vs. HTTP/HTTPS

Redundancy is a crucial principle that adds reliability to a network by providing backup systems and pathways. It ensures continuous service even if one part of the network fails. In contrast, HTTP/HTTPS protocols govern the communication of web content between clients (such as browsers) and servers. HTTPS, an encrypted version of HTTP, ensures secure data exchange over the network. While redundancy doesn't directly interact with HTTP/HTTPS, these protocols rely on stable and fault-tolerant network paths, which redundancy ensures, to maintain uninterrupted service.

Scalability vs. DNS

Scalability is the networking principle that allows systems to expand and accommodate growing data traffic, devices, or applications. It ensures that as demands increase, the network can adapt without compromising performance. The DNS protocol plays an important role in scalability by translating human-readable domain names (like www.example.com) into machine-readable IP addresses. This simplifies the management of large-scale networks and makes it easier to scale, as new devices and services can be added with minimal disruption to the system.

Security vs. DHCP

Security is an essential principle that protects data, devices, and networks from unauthorized access and potential threats. It encompasses measures like encryption, firewalls, and authentication. DHCP, or Dynamic Host Configuration Protocol, automates the process of assigning IP addresses to devices on a network, reducing manual configuration. While DHCP doesn't directly enforce security, it plays an indirect role in ensuring that only authorized devices can connect to the network, especially when combined with other security measures like network access control.

Performance Optimization vs. ARP

Performance optimization focuses on enhancing network efficiency by minimizing latency, maximizing throughput, and ensuring reliable service through techniques like Quality of Service (QoS) and traffic management. ARP, or Address Resolution Protocol, resolves the mapping between IP addresses and MAC addresses within a local network, enabling devices to communicate directly with each other. Though ARP does not directly optimize network performance, its role in facilitating efficient communication within local networks supports overall network efficiency and reliability.

4.Describe the operational principles of key networking hardware like switches, routers, access points, and firewalls. Include different servers type suitable for Mini Data Centre

Networking Devices

1. Switches

Switches are important devices that work at the data link layer (Layer 2) of the OSI model. They connect devices in the same local network and use MAC addresses to send data packets. Switches help reduce network traffic by creating separate areas where data can move, making the network work better. Managed switches offer more control with features like VLANs, which allow for better organization of the network. In the Colombo Fort office, switches can connect computers on each floor, making communication and sharing resources easier.

In the Colombo Fort office, switches can interconnect workstations on each floor, creating a reliable internal communication network.

2. Routers

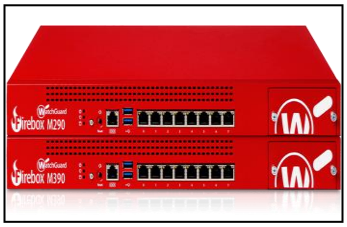
Routers operate at the network layer (Layer 3) to connect various networks and guide data packets using IP addresses. They use routing tables and protocols like OSPF and BGP to select the best routes for data to travel. This is crucial for connecting internal office networks and ensuring safe internet access. For instance, a router in a small data center can effectively manage traffic between administrative offices and IT training rooms.

A router in the Mini Data Centre directs traffic between the administrative and IT training labs while providing secure internet access.

3. Access Points (APs)

Access Points (APs) extend wireless coverage across the building, operating at the physical and data link layers. Modern APs leverage technologies like MU-MIMO to handle multiple devices simultaneously, ensuring high-speed connectivity. These devices are crucial for providing Wi-Fi access throughout the five-story Colombo Fort office, supporting employees' mobility and flexible working. APs can be strategically placed in the Colombo Fort office to provide seamless Wi-Fi access on all floors.

4. Firewalls



Firewalls protect networks by controlling incoming and outgoing traffic using set rules. They work at different levels and can be simple or very advanced. Some firewalls, like the Mini Data Centre firewall, can block unwanted access to important servers while letting in safe communication. A firewall in the Mini Data Centre can restrict access to HR servers, ensuring compliance with data protection requirements.

Server Types for a Mini Data Centre

1. Tower Servers

Tower servers operate as standalone units, much like a traditional desktop PC, with their own power supply, cooling, and storage. They are designed for independent operation, making them easy to deploy and manage without the need for extensive infrastructure. These servers rely on internal fans for cooling, allowing them to function in environments without specialized cooling systems. Tower servers are best suited for mini data centers in small businesses or setups with minimal IT needs. They excel in handling single-instance applications like file sharing, email hosting, or lightweight databases but are less scalable compared to other server types.

2. Rack Servers

Rack servers are built to fit into standardized 19-inch racks, allowing for centralized deployment and efficient use of space. They are designed for scalability, as multiple rack servers can be added to the same rack to support growing workloads. These servers are managed through remote management tools like IPMI, iLO, or DRAC, making them highly convenient for centralized IT operations. Cooling is provided externally to the entire rack, often using dedicated air or liquid cooling systems. Rack servers are ideal for mini data centers looking to maximize space while supporting diverse workloads such as web hosting, virtualization, and database management.

3. Blade Servers

Blade servers consist of thin, modular units housed in a shared chassis, which provides power, cooling, and networking for multiple blades. These servers maximize density and performance, making them suitable for high-demand environments with limited physical space. The chassis centralizes management, reducing cabling complexity and simplifying system updates. Blade servers are well-suited for mini data centers requiring high performance, scalability, and efficiency. They excel in workloads like high-performance computing, virtualization, and cloud services but require a higher initial investment and specialized infrastructure.

4. Hyper-Converged Infrastructure (HCI) Servers

Hyper-converged infrastructure servers integrate compute, storage, and networking into a single unit, simplifying deployment and management. They are designed to operate in a virtualized environment, where resources are pooled and workloads are distributed across nodes in a cluster. This architecture provides high scalability and resiliency, making HCI servers ideal for data-intensive applications, private clouds, and virtual desktop infrastructure (VDI). For mini data centers, HCI servers offer a streamlined solution with centralized management tools, though the initial setup cost may be higher compared to traditional server setups.

5. Edge Servers

Edge servers are compact and designed to process data closer to the source, reducing latency and bandwidth usage. They are often deployed in remote locations or environments with specific IoT or real-time processing needs. These servers are rugged and capable of operating in challenging conditions, making them suitable for decentralized operations. For a mini data center, edge servers are useful when there’s a need to handle localized workloads such as IoT data processing or remote site analytics, ensuring quick response times and efficient resource utilization.

5.Identify and explain different networking software needed for this project, including OS and applications, and discuss their interdependence with hardware.

1.Operating Systems (OS)

a. Operating Systems (OS)

Workstation Operating Systems

Windows 10/11 Professional/Enterprise or Linux (Ubuntu/CentOS)

Purpose

These systems are needed for the computers used by workers in different areas, like finance, IT, office work, and management. The computers must have a reliable and safe system that works with both wired and wireless internet, checks who is using the computer, and lets people access shared things like files, printers, and servers.

Interdependence with Hardware

The operating system (OS) works directly with the computer's hardware, controlling things like the CPU, memory, storage, and network connections (like Ethernet and Wi-Fi). The computer's hardware, like its memory and processing speed, affects how well the OS and the programs on it run.

b. Server Operating Systems

Windows Server 2019/2022 or Linux Server (Ubuntu Server/CentOS)

Purpose

The computers that store files, run applications, and keep backups need a dependable and safe operating system to handle various tasks, like managing internet connections, domain names, user accounts, sharing files and printers, and saving data.

Interdependence with Hardware

Server hardware must be robust with sufficient storage (e.g., SSDs/HDDs for large data sets), high-speed processors, and RAM to handle the demands of running multiple virtual machines, database services, and file-sharing operations. The server OS interacts with this hardware to ensure efficient resource allocation and load balancing.

2.Network Operating Systems and Management Software

a. Network Management Software

SolarWinds Network Performance Monitor, Nagios, PRTG Network Monitor

Purpose

These tools help monitor the performance of network hardware (routers, switches, firewalls) and connected devices, track bandwidth usage, detect outages, and perform diagnostics on the network.

Interdependence with Hardware

The network management software relies on routers, switches, and firewalls to collect data and provide insights into network performance. The proper configuration and uptime of hardware are essential for the software to effectively monitor and analyze the network's health.

b. Switch and Router OS

Cisco IOS, Juniper Junos OS, or Arista EOS

Purpose

These are the software platforms running on network hardware such as routers and switches. They provide networking features such as routing, VLANs, traffic filtering, and network segmentation.

Interdependence with Hardware

The network hardware (routers, switches) is responsible for forwarding traffic and managing data flow between workstations and servers. The software on these devices defines how data is routed, prioritized, and secured across the network. The hardware's processing power, interface types, and speed must align with the requirements of the router/switch OS to prevent bottlenecks.

3.Network Security Software

a. Firewall Software

pfSense, Cisco ASA, FortiGate Firewall

Purpose

These firewalls control incoming and outgoing network traffic based on security rules, protecting the network from unauthorized access and cyber threats.

Interdependence with Hardware

The firewall software relies on dedicated firewall hardware (or virtual appliances) that have sufficient throughput and security processing capabilities to enforce security policies. The hardware must support the firewall's features, such as deep packet inspection and VPN tunnels, without compromising performance.

b. Antivirus and Endpoint Security

McAfee, Sophos, or Symantec Endpoint Protection

Purpose

These software solutions are necessary to protect the workstations, servers, and other network devices from malware, viruses, and security breaches.

Interdependence with Hardware

The security software works with the hardware's CPU and memory to scan files and processes for malicious activity in real time. A high-performance CPU and sufficient RAM will improve the software's ability to handle multiple scans simultaneously without causing system slowdowns.

c. VPN (Virtual Private Network) Software

OpenVPN, Cisco AnyConnect, Palo Alto GlobalProtect

Purpose

VPN software enables secure remote access to the network, allowing employees working remotely to connect to the office network securely.

Interdependence with Hardware

VPN software requires sufficient network bandwidth and encryption/decryption capabilities from the hardware to ensure secure and efficient data transmission. Routers and network interfaces must support VPN protocols, and workstations need enough CPU power to handle encryption without affecting performance.

4. Data Backup and Storage Software

a. Backup Software

Veeam, Acronis, or Backup Exec

Purpose

Backup software is crucial for creating backups of critical data, applications, and server configurations, ensuring business continuity in case of hardware failure or disaster.

Interdependence with Hardware

Backup solutions rely on the data storage hardware (e.g., SAN or NAS devices, RAID arrays) to store backup copies. The backup software communicates with this storage hardware to schedule and execute backup tasks, ensuring data is stored securely.

5. Collaboration and Productivity Software

a. Office Suite

Microsoft Office 365, Google Workspace

Purpose

These applications enable employees to perform word processing, spreadsheet analysis, email, and collaboration tasks. Cloud-based solutions like Office 365 can be used to integrate with internal servers for file sharing and collaboration.

Interdependence with Hardware

These applications interact with the workstation hardware (processor, RAM, storage) to run smoothly. A higher-end workstation hardware configuration may be needed for tasks requiring heavy computations or large datasets, like financial analysis.

b. Email Server Software

Microsoft Exchange, Zimbra, or Google Workspace for Business

Purpose

Email servers handle the sending, receiving, and storing of email messages within the network.

Interdependence with Hardware

The server hardware used for email hosting must support the continuous running of the email service, which includes ensuring high disk I/O for message storage and sufficient CPU and RAM to handle email traffic volumes.

6.Explore potential server types to handle file storage, email, and web hosting, considering cost and performance to make a final recommendation.

1. File Storage: Network Attached Storage (NAS)

Recommendation: A mid-range NAS device, such as Synology DiskStation DS920+ or QNAP TS-453D.

Why This Option?

- NAS devices are more affordable than traditional file servers, with prices significantly lower for the same storage capacity.
- They are simple to deploy and maintain, reducing the need for highly skilled IT personnel.
- Supports shared access for 250 employees, enabling secure file storage and sharing.
- Drives can be added later as storage needs grow, which aligns with budget constraints.
- Lower power consumption compared to traditional servers saves operational costs.

For file storage, a Network Attached Storage (NAS) device, such as the Synology DiskStation DS920+ or QNAP TS-453D, is the ideal solution for the Colombo Fort office building. NAS devices are highly cost-effective, offering centralized file storage and easy access for all 250 employees across the five floors of the building. With built-in RAID configurations, they ensure data redundancy, protecting the company’s critical files. The scalability of NAS devices aligns with the company’s future growth plans, allowing additional storage to be added as required. These devices are compact and energy-efficient, making them well-suited to the on-site data center on the 5th floor without burdening the power and cooling infrastructure.

2. Email Hosting: Cloud-Based Solution (Microsoft 365 or Google Workspace)

Recommendation: Microsoft 365 Business Basic or Google Workspace Business Starter for cloud email hosting.

Why This Option?

- Avoids the high cost of purchasing and maintaining an on-premises email server, which includes hardware, software licenses, and IT staff for management.
- Cloud solutions offer high uptime (99.9%) and scalability to accommodate growth without additional hardware investment.
- Includes spam filtering, encryption, and other security measures, which would otherwise require separate purchases for an on-premises setup.

- Both platforms ensure data security and compliance with international standards, critical for a financial services firm.

For email hosting, a cloud-based solution such as Microsoft 365 Business Basic or Google Workspace Business Starter is recommended. This eliminates the need for physical email servers, saving space in the 5th-floor data center. Cloud hosting ensures seamless email communication among employees spread across different floors and departments, supporting efficient workflows in the Colombo Fort office. High uptime guarantees uninterrupted access, critical for the financial services firm's operations. Additionally, these platforms provide built-in security features, such as encryption and spam protection, which are essential for handling sensitive client data while complying with financial regulations.

3. Web Hosting Virtual Private Server (VPS)

Recommendation: A VPS from a reliable provider such as Linode, DigitalOcean, or AWS Lightsail.

Why This Option?

- VPS hosting eliminates the need for dedicated hardware for web hosting, offering a cost-effective monthly subscription.
- Suitable for hosting intranet sites, internal applications, or customer-facing websites with moderate traffic.
- Resources (CPU, RAM, storage) can be upgraded as needed without additional hardware purchases.
- VPS services are hosted in data centers with robust power and cooling infrastructure, ensuring consistent uptime.

For web hosting, a Virtual Private Server (VPS) hosted by providers like Linode, DigitalOcean, or AWS Lightsail is the most suitable choice for this commercial office. A VPS eliminates the need for dedicated hardware on-site, freeing up physical space and reducing power consumption in the data center. This option provides sufficient resources for hosting internal applications, intranet websites, or public-facing portals that financial analysts, IT support staff, and other employees need to access. Its flexibility allows for resource scaling, which is crucial as the firm expands its digital footprint.

4. Application Hosting and Backup: On-Premises Hybrid Server

Recommendation: Entry-level rack-mounted server such as Dell PowerEdge R240 or HPE ProLiant DL20 Gen10.

Why This Option?

- On-premises servers ensure sensitive financial and HR data is kept locally, meeting compliance requirements and minimizing risks of data breaches.
- Entry-level servers provide sufficient processing power and storage for hosting internal applications like financial tools, HR systems, and backup solutions.
- These servers can host multiple virtual machines, optimizing hardware utilization and saving costs on additional physical servers.
- On-site backup servers allow quick recovery in case of hardware failures or cyber incidents, essential for the financial services industry.

For application hosting and backup, an entry-level on-premises hybrid server, such as the Dell PowerEdge R240 or HPE ProLiant DL20 Gen10, is recommended. These servers will be housed in the secure area of the 5th-floor data center, ensuring critical financial and HR data remain on-site as per compliance requirements. The hybrid server provides the computing power needed for hosting internal applications used by financial analysts, IT training labs, and HR teams. Additionally, it supports local data backups, allowing quick recovery in case of system failures. Its compact design and energy-efficient operation make it a cost-effective choice for the Colombo Fort office’s infrastructure.

7.Critically evaluate the topology protocol selected for this project and demonstrate the efficient utilization of a networking system.

What is the most suitable topology | Why picked it

I chose the star topology for this task because it provides centralized control, making it easier to manage access rules and network segmentation, which are crucial for this scenario. Each department on different floors has specific access needs, such as restricting financial staff from accessing HR data while allowing Executive Managers to access all networks. The star topology allows these rules to be implemented efficiently using VLANs and Access Control Lists (ACLs). Additionally, it ensures fault isolation, if one connection fails, it won't affect others, and supports scalability, allowing new users or networks to be added easily. While it has a risk of a single point of failure, this can be mitigated with redundancy, making it a reliable and flexible choice for the project.

Advantages of the Star Topology

The star topology is highly efficient for managing access controls, which is must in this case. For example, the 1st floor (financial department) requires a dedicated network that cannot be accessed by other departments. With a star topology, this can be achieved using VLANs (Virtual Local Area Networks) configured on the central switch. VLANs logically separate the financial network from others, ensuring sensitive financial data remains protected.

Similarly, the 2nd floor (IT support) can have its network segregated while still maintaining controlled access to the financial network for troubleshooting purposes. This is possible because VLANs allow selective communication between networks. For example, IT support staff can access financial systems for maintenance, but financial staff cannot access IT systems.

The 3rd floor (HR and Executive Managers) benefits from the centralized control of the star topology. Executive Managers, who need access to all networks, can be given permissions at the central switch level. For instance, while no one can access the Executive Manager network, the central switch allows Executive Managers to access both the HR and IT support networks, fulfilling their unique requirements without compromising security.

Another benefit is fault isolation. If a single cable or device fails, it does not impact the rest of the network. This feature is crucial for maintaining uninterrupted services for essential areas like financial operations or IT support. Additionally, the topology supports scalability, meaning that new devices or departments can be added easily by connecting them to the central switch without affecting the rest of the system.

Challenges and Limitations

While the star topology offers robust control, it has a critical drawback, the single point of failure. In this case, the entire building's network depends on the central switch. If it fails, connectivity to the IT training labs on the 4th floor and the main server room on the 5th floor will also be disrupted, halting all operations. For example, if the switch malfunctions, Executive Managers will lose their ability to access IT training resources or manage HR tasks, which could cause operational delays.

This risk can be mitigated by introducing redundancy. For instance, adding a secondary switch as a backup ensures network continuity. If the primary switch fails, the backup can immediately take over, maintaining access to critical systems like the server room and HR network.

Another issue is cost and resource utilization. A central switch capable of handling multiple VLANs, enforcing access control, and supporting high-speed communication can be expensive. Additionally, the structured cabling required to connect all devices to the central switch may significantly increase the installation cost, especially for a multi-floor building like this commercial office building located in Colombo Fort.

Real-World Implementation

In this case, the 4th floor IT training labs require a separate network for training purposes. With the star topology, this can be easily set up as an isolated VLAN. However, since Executive Managers need access to this network, permissions can be granted at the central switch level. This allows seamless connectivity while keeping the training network isolated from other departments, such as finance or HR

The 5th floor server room must remain secure, as it houses critical data. By configuring VLANs and ACLs, only authorized IT support and Executive Managers can access this network, ensuring that sensitive information is not exposed to unauthorized users.

Access Control Policies

Efficient utilization requires robust access control policies to prevent unauthorized access. For example,

- Implement ACLs on the central switch to enforce the rules outlined in the scenario. For instance, financial and IT support staff cannot access HR or Executive Manager networks, while Executive Managers have unrestricted access.

- Use port-based network access control to ensure only authenticated devices and users can connect to the network.

These measures enhance security, ensuring that network resources are used only by those with legitimate access.

Bandwidth and Traffic Management

Efficient utilization involves managing network traffic to prioritize critical operations and prevent congestion. Strategies include

- Configure QoS to prioritize essential services. For instance, financial transactions on the 1st floor can be given higher priority than training activities on the 4th floor.

- Tools like SNMP (Simple Network Management Protocol) or NetFlow can monitor traffic patterns and detect potential issues, such as excessive bandwidth use in the IT training labs.

By managing traffic effectively, the network ensures smooth operation even during peak usage.

Comparison of Network Topologies for this Task

Feature	Star Topology	Bus Topology	Ring Topology	Mesh Topology
Ease of Implementation	Easy to set up with centralized control, suitable for managing VLANs and access rules.	Simple to implement but difficult to manage access restrictions for specific departments.	Moderate complexity, requires configuring token passing for data access, which is not flexible for specific tasks.	Complex and time-consuming due to interconnecting multiple devices, overkill for a scenario with clear hierarchies.
Scalability	Highly scalable, new devices or departments can be easily added via the central switch.	Limited scalability, adding new devices disrupts the network and requires rewiring.	Difficult to scale, adding devices requires modifying the entire ring.	Highly scalable but requires significant resources to connect all devices, making it costly for this task.
Fault Isolation	Easy to identify and isolate faults, a single device failure doesn't impact the entire network.	Failure of the main bus affects all devices, making it less reliable.	A failure in one device or connection can disrupt the entire ring unless a dual-ring configuration is used.	Fault isolation is strong, but the complexity of diagnosing issues in a large network can slow resolution.
Access Control	Centralized control with VLANs and ACLs ensures access rules (e.g. HR, financial) are well-enforced.	No central control, enforcing access restrictions for departments is challenging.	Limited flexibility for enforcing specific access rules, token-passing mechanisms don't align with task needs.	High flexibility in access control, but the complexity of managing multiple connections may outweigh the benefits.
Cost	Cost-effective for medium-sized networks, requires one central switch and cabling for	Low initial cost but becomes inefficient for larger networks.	Moderate cost, requires multiple connections but is less efficient for department-	High cost due to extensive cabling and hardware requirements, which is unnecessary for

	each device.		level access needs.	this task.
Performance	High performance, dedicated lines for each connection prevent congestion.	Performance degrades as more devices share the same bus.	Moderate performance, token passing introduces latency, especially for large networks.	Excellent performance but unnecessary for a network with limited interdepartmental traffic.
Security	Strong security with VLANs and ACLs to segment traffic and limit access to sensitive data.	Poor security, data travels through the same bus, making it vulnerable to eavesdropping.	Moderate security, token-passing mechanisms offer some isolation but lack granular access control.	Strong security but excessive complexity for the task, making management cumbersome.
Suitability for Task	Best suited for this task due to centralized management, scalability, and ability to enforce access rules.	Unsuitable as it cannot handle the strict access requirements and risks downtime due to a single cable failure.	Unsuitable as it cannot enforce department-specific access rules and is prone to disruptions.	Overly complex and expensive for the specific access and control requirements in this scenario.